

Homework 1

- 1 Solve the spacelike geodesic equation for AdS_{d+1} in Poincaré coordinates. In particular, write down the leading order terms for the proper length between two points near the boundary $z = \epsilon$ with a coordinate separation in a spatial direction, say $x_2 - x_1 = L$.

- 2 Consider the world-line action of a point particle

$$S = \frac{1}{2} \int d\tau (\eta^{-1} \dot{X}^\mu \dot{X}_\mu - \eta m^2). \quad (1)$$

How does η transform under the reparameterization of the parameter τ ? Show that it is classically equivalent to the action

$$\tilde{S} = -m \int d\tau (-g_{\alpha\beta} \dot{X}^\alpha \dot{X}^\beta)^{1/2}. \quad (2)$$

- 3 Work out the Ricci tensor for the metric

$$ds^2 = -f(r)dt^2 + h(r)dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2) \quad (3)$$

- 4 Derive the equations of motion for the Jackiw-Teitelboim gravity, a two dimensional theory of gravity coupled with dilaton

$$S = \frac{1}{16\pi G} \int dx^2 \sqrt{-g} \Phi (R + 2) \quad (4)$$

where Φ is a dilaton field.